

END SEMESTER ASSESSMENTS, SPRING 2024

Program: B.TECH COMPUTER SCIENCE & BUSINESS SYSTEMS/ B.TECH COMPUTER SCIENCE & ENGINEERING/B.TECH COMPUTER SCIENCE & ENGINEERING (LATERAL)/B.TECH COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)/B.TECH COMPUTER SCIENCE AND ENGINEERING (IOT, CYBERSECURITY AND BLOCKCHAIN TECHNOLOGY)	
Subject Name: OPERATIONS RESEARCH/OPTIMIZATION TECHNIQUE	
Subject Code: 7000023101/7000013101/7000033101/ 7000103101/7000113101	Semester: IV
Duration: 2 Hours	Maximum Marks: 50

Note: Answer the following questions as directed.

Group A**(Answer All questions)****(Each question carries 1 mark)**

1. Write the correct answer from the options given: (10 X 1)

- (i) The number of basic solutions of a system of m linear equations in n variables ($m < n$) is
- (a) $n - m$ (b) $m + n$
(c) mn (d) $n_c m$
- (ii) In simplex method, if all the entries in the pivot column are negative or zero, then
- (a) The solution is infeasible (b) There is alternative optima
(c) The solution is unbounded (d) The solution is degenerate
- (iii) The maximum value of $Z = 3x - y$, subject to $x + y \leq 1$, and $x, y \geq 0$ is
- (a) 0 (b) 2
(c) 3 (d) -1
- (iv) When the total allocations in a transportation model of $m \times n$ size do not equal to $m + n - 1$, the situation is known as
- (a) Unbalanced situation (b) Tie situation
(c) Degeneracy (d) None of these

- (v) The coefficient of artificial variable in the objective function (Maximization) is
 (a) 0 (b) M
 (c) $-M$ (d) 1
- (vi) Which of the following methods is used to verify the optimality of the current solution of the transportation problem?
 (a) Least cost method (b) North-west corner method
 (c) Vogel's approximation method (d) MODI method
- (vii) In an assignment problem involving 5 workers and 5 jobs, the total number of possible assignments is
 (a) 5 (b) 120
 (c) 20 (d) 25
- (viii) PERT is used in the preparation of
 (a) Budgeting (b) Evaluating
 (c) Scheduling (d) Finalizing
- (ix) The term 'Jokeying' in queuing theory refers to
 (a) Shifting from one queue to another parallel queue (b) Not entering the long queue
 (c) Leaving the queue (d) None of these
- (x) The set of vectors $\{(1, 0, 0), (0, 1, 1), (1, 2, 3)\}$ is
 (a) Linearly dependent (b) Linearly independent
 (c) Both (a) and (b) (d) None of these

Group B

(Answer any Four questions out of Six)

(Each question carries 5 marks)

(4 X 5)

2. A manufacturer of patent medicines is preparing a production plan on medicines A and B. There are sufficient ingredients available to make 20000 bottles of A and 40000 bottles of B, but there are only 45000 bottles into which either of the medicines can be put. Furthermore, it takes three hours to prepare enough material to fill 1000 bottles of A, it

takes one hour to prepare enough material to fill 1000 bottles of B and there are 66 hours available for this operation. The profit is Rs. 8.00 per bottle of A and Rs. 7.00 per bottle of B. Formulate the problem as a linear programming problem to maximize the profit.

3. Find the basic feasible solutions of the system of equations:

$$x_1 + x_2 + x_3 = 8,$$

$$3x_1 + 2x_2 = 18,$$

$$x_1, x_2, x_3 \geq 0.$$

4. Solve the following L.P.P. by simplex method:

$$\text{Maximize: } Z = 3x_1 + 2x_2$$

Subject to

$$2x_1 + x_2 \leq 2,$$

$$3x_1 + 4x_2 \geq 12,$$

$$\text{and } x_1, x_2 \geq 0.$$

5. Use VAM to find the basic feasible solution of the following transportation problem:

	D_1	D_2	D_3	D_4	a_i
O_1	21	16	25	13	11
O_2	17	18	14	23	13
O_3	32	27	18	41	19
b_j	6	10	12	15	

6. Listed in the table are the activities and sequencing necessary for a maintenance job on the heat exchangers in a refinery.

Activity	Description	Predecessor Activity
A	Dismantle pipe connections	-
B	Dismantle heater, closure, and floating front	A
C	Remove tube bundle	B
D	Clean bolts	B
E	Clean heater and floating head front	B
F	Clean tube bundle	C
G	Clean shell	C
H	Replace tube bundle	F, G
I	Prepare shell pressure test	D, E, H
J	Prepare tube pressure test and reassemble	I

Draw a network diagram of activities for the project.

7. A particular item has a demand of 9,000 units/year. The cost of each procurement is Rs.100 and the holding cost per unit is Rs.2.40 per year. The replenishment is instantaneous and no shortages are allowed. Determine:-

- The economic lot size,
- The number of orders per year,
- The total cost per year, if the cost of one unit is Rs.1.

Group C

(Answer any Two questions out of Four)
(Each question carries 10 marks)

(2 X 10)

8. Use Charnes Big-M method to solve the following L.P.P.:

$$\text{Maximize } Z = -2x_1 - x_2$$

Subject to

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$\text{and } x_1, x_2 \geq 0.$$

9. Solve the LPP by dual simplex method:

$$\text{Maximize } z = 10x_1 + 6x_2 + 2x_3$$

Subject to

$$-x_1 + x_2 + x_3 \geq 1$$

$$3x_1 + x_2 - x_3 \geq 2$$

$$x_1, x_2, x_3 \geq 0.$$

10. Find the optimal solution of the assignment problem with the following cost matrix using Hungarian method:

Job Men	I	II	III	IV	V
A	10	5	13	15	16
B	3	9	18	13	6
C	10	7	2	2	2
D	7	11	9	7	12
E	7	9	10	4	12

1. A self-service store employs one cashier at its counter. 9 customers arrive in an average every 5 minutes while the cashier can serve 10 customers in 5 minutes. Assuming Poisson distribution for arrival rate and exponential distribution for service time, find the followings:

- Average number of customers in the system
- Average number of customers in the queue
- Average time a customer spends in the system
- Average time a customer waits before being served.